

# Biological Safety Manual - Chapter 10: Decontamination and Disinfection

## Summary

This chapter describes basic strategies for decontaminating surfaces, items, and areas in laboratories to eliminate the possibility of transmission of infectious agents to laboratory workers, the general public, and the environment. Factors necessary for environmentally mediated infection transmission are reviewed as well as methods for sterilization and disinfection and the levels of antimicrobial activity associated with liquid chemical germicides.

## Body

## Title

Biological Safety Manual - Chapter 10: Decontamination and Disinfection

## Introduction

This chapter describes basic strategies for decontamination or disinfection of environmental surfaces, items, and areas in laboratories and other practices to mitigate the possibility of transmission of infectious agents to laboratory workers, the public, and the environment. Regulatory oversight, terminology, factors necessary for environmentally-mediated transmission of infection (e.g., aerosol generation, contact, indirect contact), methods for sterilization and disinfection, and the levels of antimicrobial activity associated with liquid chemical disinfectants are reviewed in this appendix. One must remember that aerosol-generating procedures should be conducted in containment. Accidents involving infectious aerosols have been a source of contamination within the laboratory setting and may impact the method chosen for decontamination. General approaches are emphasized instead of detailed protocols and methods. It is important to follow the manufacturer's instructions for use when performing decontamination practices in the laboratory.

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# I. Antimicrobial Products - U.S. Regulations

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Antimicrobial pesticides (e.g., disinfectants) are classified as pesticides and are regulated by both the United States Environmental Protection Agency (EPA) under the authority of the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA)<sup>1,2</sup> and the United States Food and Drug Administration (FDA), Center for Devices and Radiologic Health by the Food Quality Protection Act (FQPA).<sup>3</sup> The laboratory is responsible for selecting an appropriate EPA-registered product and using it according to the manufacturer's instructions on the product label. The more commonly used public health antimicrobial products are described in the Glossary (e.g., sporicides, disinfectants, and sanitizers). The list of selected EPA-registered disinfectants are available on the EPA website.

The FDA has defined three types of liquid chemical germicides for processing medical devices, and these germicides are regulated as auxiliary devices (FDA 1977 Policy Manual):

1. sterilant/high-level disinfectant;
2. intermediate-level disinfectant; and
3. low-level disinfectant.

Disinfectants used in the laboratory include those recommended by equipment manufacturers and a broad-spectrum product, typically an intermediate-level disinfectant (i.e., a product with a mycobacteriology claim). Safe use of chemicals within the laboratory falls under the OSHA Laboratory Standard.<sup>4</sup>

## II. Environmentally Mediated Infection Transmission

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Laboratory-associated infections (LAIs) can be transmitted directly or indirectly from environmental sources within the laboratory (e.g., air, fomites and laboratory instruments, aerosols, and splashes) to laboratory staff. Fortunately, LAI are relatively rare events because there are several requirements necessary for environmental transmission to occur.<sup>5,6</sup> The requirements needed for environmental transmission include:

1. the presence of a pathogen of sufficient virulence,
2. sufficient dose of a pathogen to cause infection (i.e., infectious dose),
3. a mechanism of transmission of the pathogen from environment to the host,
4. a correct portal of entry to a susceptible host, and
5. the immune status of the host.

To accomplish successful transmission from an environmental source, all of these requirements for the chain of infection must be present. The absence of any one element will reduce and/or prevent the potential for transmission. Additionally, the pathogen in question must overcome environmental stresses to retain viability (e.g., ability to form biofilms in low, nutrient-moist environments or distribution systems; ability to survive dehydration), virulence, and the capability to initiate infection in the host. In the laboratory setting, high concentrations of pathogens are commonplace, and containment of environmental surfaces (e.g., benchtops, equipment, personal protective equipment (PPE)) and hands of the laboratorian may occur. Aerosol generation procedures and those that generate splashes may also contaminate surfaces, personnel, and potentially expose workers (e.g., inhalation, contact with mucous membranes) to pathogens. Reduction of environmental microbial contamination by both containment (e.g., performing aerosol-generating procedures in a biological safety cabinet or glove box) and conventional cleaning procedures is often sufficient to reduce, but not eliminate, the risk of environmentally mediated transmission. It is the general practice in laboratories to use both cleaning and surface disinfection or sterilization procedures to mitigate the potential for infection transmission. In addition, proper hand hygiene and appropriate use of PPE (e.g., gloves, lab coat/smock, safety glasses, goggles, respirators) are also important factors in preventing LAIs.

## III. Principles of Cleaning, Disinfection, and Sterilization

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In order to implement a laboratory biosafety program it is important to understand the principles of decontamination, cleaning, sterilization, and disinfection. We review here the definitions of sterilization, disinfection, antisepsis, decontamination, and sanitization to avoid misuse and confusion. The definitions and implied capabilities of each inactivation procedure are discussed with an emphasis on achievement and, in some cases, monitoring of each state.

### Cleaning

Cleaning is the removal of gross contamination from a surface to the extent necessary for further processing for intended use. In these cases, cleaning can be used to remove microorganisms and other associated contaminants (e.g., blood, tissues, culture media) from a surface by physical means but may not provide any antimicrobial activity. Cleaning is often an essential pre-requisite to disinfection or sterilization processes to ensure the optimal activity of the antimicrobial effects of disinfectants or sterilization processes. Biofilms may be present in the laboratory (e.g., sinks, plumbing fixtures, fluid-filled lines of laboratory equipment, water containing reservoirs, incubator humidification systems) and are often difficult to treat/disinfect. Most biofilms require physical cleaning (e.g., scrubbing) and the use of compatible oxidative disinfectants (e.g., chlorine dioxide, peroxyacetic acid, ozone). In some situations, replacing tubing and distribution lines may be necessary.

### Disinfection

Disinfection is generally a less lethal process than sterilization; it eliminates nearly all recognized pathogenic microorganisms, but not necessarily all microbial forms (e.g., bacterial spores) present on inanimate objects. Disinfection does not ensure a kill level and lacks the margin of safety achieved by sterilization procedures. The effectiveness of a disinfection procedure is controlled by several factors, each of which may have a pronounced effect on the end results. Factors affecting disinfection include the following:

- 1. Nature and number of contaminating microorganisms (especially the presence of bacterial spores);
- 2. Amount of organic matter present (e.g., soil, feces, and blood);
- 3. Type and condition of surfaces, instruments, devices, and materials to be disinfected;
- 4. Temperature; and
- 5. Contact (exposure) time.

By definition, chemical disinfection, especially high-level disinfection differs from chemical sterilization by the lack of sporicidal power. This is an oversimplification of reality because a few chemical disinfectants do kill large numbers of spores even though high concentrations and several hours of exposure may be required. Non-sporicidal disinfectants may differ in their capacity to accomplish disinfection or decontamination. Some disinfectants rapidly kill only the ordinary vegetative forms of bacteria such as staphylococci and streptococci, some forms of fungi, and lipid-containing viruses, whereas others are effective against such relatively resistant organisms as *Mycobacterium bovis* or *Mycobacterium terrae*, non-enveloped viruses, and most forms of fungi.<sup>9</sup>

In general, most laboratories use a disinfectant that has a broad range of activity; thus, most labs should select a product with a tuberculocidal/mycobactericidal claim for routine purposes. Many of these products will also have claims that meet the OSHA Bloodborne Pathogens Standard.<sup>10,11</sup>

## Sterilization

Any item, device, or solution is sterile when it is completely free of all forms of living microorganisms, including spores and viruses. The definition is categorical and absolute; an item is either sterile or it is not. Sterilization can be accomplished by dry or moist heat gases and vapors (e.g., chlorine dioxide, ethylene oxide, formaldehyde, hydrogen peroxide, methyl bromide, nitrogen dioxide, ozone, propylene oxide) plasma sterilization technology, and radiation (e.g., gamma, e-beam in industry).

From an operational standpoint, a sterilization procedure cannot be categorically defined because the likelihood that an individual microorganism survives is never zero. Rather, the procedure is defined as a process, after which the probability of a microorganism surviving on an item subjected to treatment is less than one in one million. This is referred to as a sterility assurance level (SAL) of 10<sup>-6</sup>.<sup>12-14</sup> Laboratories use sterilization techniques for producing media, sterilizing glassware and other items, and decontaminating waste.

## IV. Decontamination

Decontamination renders an area, device, item, or material safe to handle in the context of being reasonably free from a risk of disease transmission. The primary objective of decontamination is to reduce the level of microbial contamination so that transmission of infection is prevented. The decontamination process may involve the cleaning of an instrument, device, or area with ordinary soap and water. In laboratory settings, decontamination of items, used laboratory materials, and regulated laboratory wastes is often accomplished by a sterilization procedure such as steam autoclaving, which may be the most cost-effective way to decontaminate a device or an item.

The presence of any organic matter necessitates longer contact time with a decontamination method if the item or area is not pre-cleaned. For example, a steam cycle used to sterilize pre-cleaned items can be 20 minutes at 121°C. When steam sterilization is used to decontaminate laboratory waste that contains items that have a high bio-burden and there is no pre-cleaning (i.e., infectious waste), the cycle times are generally longer and should be verified and validated for the typical load. Validation involves the combined use of thermocouples and biological indicators (BIs) placed throughout the load to ensure penetration of steam into the waste. Verification can be accomplished by routine monitoring of the steam sterilization cycles (i.e., cycle times, pressure, temperature) and by placing BIs within the load.<sup>15</sup> In addition to time, temperature may also be increased to ensure inactivation of pathogens.<sup>16-18</sup> Decontamination in laboratory settings often requires longer exposure times because pathogenic microorganisms may be protected from contact with steam.

Chemical disinfectants used for decontamination range in activity from high-level disinfectants (e.g., high concentrations of sodium hypochlorite [chlorine bleach]), which might be used to decontaminate spills of cultured or concentrated infectious agents in research or clinical laboratories, to low-level disinfectants or sanitizers for general housekeeping purposes or spot decontamination of environmental surfaces in healthcare settings. Resistance of selected organisms to decontamination is presented in descending order in Figure 1. If dangerous and highly infectious agents are present in a laboratory, the methods for decontamination of spills, laboratory equipment, biological safety cabinet, or infectious waste are very significant and may include prolonged autoclave cycles, incineration, or gaseous treatment of surfaces.

Figure 1: Descending Order of Relative Resistance to Disinfectant Chemicals

|                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|
| <b>Prions</b>                                                                                                      |
| <b>Bacterial Spores</b><br><i>Bacillus subtilis</i> , <i>Clostridium sporogenes</i> , <i>Clostridium difficile</i> |
| <b>Mycobacteria</b><br><i>Mycobacterium bovis</i> , <i>M. terrae</i> , and other Nontuberculous mycobacteria       |
| <b>Non-enveloped or Small Viruses</b><br>Poliovirus, Coxsackievirus, Rhinovirus                                    |
| <b>Fungi</b><br><i>Trichophyton</i> spp., <i>Cryptococcus</i> spp., <i>Candida</i> spp.                            |

|                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Vegetative Bacteria</b></p> <p><i>Pseudomonas aeruginosa</i>, <i>Staphylococcus aureus</i>, <i>Salmonella choleraesuis</i>, Enterococci</p>    |
| <p><b>Enveloped or Medium-size Viruses</b></p> <p>Herpes simplex virus, CMV, Respiratory syncytial virus, HBV, HCV, HIV, Hantavirus, Ebola virus</p> |

**Note:** There are exceptions to this list. *Pseudomonas* spp. are sensitive to high-level disinfectants. However, in biofilms, the protected cells and those within free-living amoeba, or existing as persister cells (viable but not culturable) within the biofilm, can approach the resistance of bacterial spores to the same disinfectant. The same is true for the resistance to glutaraldehyde by some nontuberculous mycobacteria, some fungal ascospores of *Microascus cinereus* and *Chaetomium globosum*, and the pink-pigmented *Methylobacteria*. Prions are also resistant to most liquid chemical germicides and are discussed in the last part of this appendix.

## A. Space Decontamination

Space decontamination is a specialized activity and should be performed by specialists with proper training and protective equipment.<sup>19-24</sup> Decontamination requirements for laboratory space have an impact on the design of these facilities. The interior surfaces of laboratories must be easy to clean and decontaminate. Penetrations in BSL-3 surfaces should be sealed or capable of being sealed for decontamination purposes. Thus, in the BSL-3 laboratory, surface decontamination, not fumigation, is the primary means of decontaminating space. Care should be taken that penetrations in the walls, floors and ceilings are kept to a minimum and are sight sealed. Verification of the seals is highly recommended, but usually not required, for BSL-3 laboratories. The BSL-4 laboratory design requires interior surfaces that are water resistant and sealed to facilitate fumigation. Periodic fumigation is required in the BSL-4 suit laboratory to allow routine maintenance and certification of equipment.

Procedures for decontamination of large spaces such as incubators or rooms are varied and influenced significantly by the type of etiologic agent involved, the characteristics of the structure containing the space, and the materials present in the space. The primary methods for space decontamination follow. Fumigants that are currently used are either gases, vapors, mists, or fogs (dry mists). Fumigants that are gases obey gas laws, can evenly distribute throughout the room, and are easily scalable by increasing the volume of gases used. Fumigants applied as mists or fogs do not behave like gases and are particles (<1–12 μ in size) that settle onto surfaces being treated.

### Paraformaldehyde and Formaldehyde Gas

Paraformaldehyde and solutions of formaldehyde have been used to generate formaldehyde gas and mists; historically, they have been used in laboratory settings for decontamination of large spaces and biological safety cabinets.<sup>25,26</sup> When using formaldehyde and paraformaldehyde, take safety precautions,<sup>27,28</sup> federal regulations, state regulations, and local regulations into consideration.<sup>29</sup> Information on environmental and safety issues related to paraformaldehyde is available on the EPA's Pesticides website.

Formaldehyde is recognized as a known human carcinogen.<sup>30</sup> There is at least one EPA-registered paraformaldehyde product available for the decontamination of laboratories. It is important that paraformaldehyde is used per labeling instructions and that a fumigation management and safety plan that meets federal, state, and local regulations is prepared in advance of application and is implemented during application. For use as a space decontamination agent, the standard concentration of formaldehyde is 0.3g/ft3 (approximately 8,000 ppm) with a relative humidity of between 60 and 85%.<sup>31</sup> Increasing the amount of paraformaldehyde is not advised, as the lower explosive limit for formaldehyde gas is 7% (70,000 ppm).<sup>32</sup> It is recommended that formaldehyde gas decontamination be performed only by highly experienced individuals.

### Hydrogen Peroxide Vapor

Hydrogen peroxide can be vaporized and used for the decontamination of glove boxes as well as small room areas. Vapor phase hydrogen peroxide has been shown to be an effective sporicide at concentrations ranging from 0.5 mg/L to <10 mg/L. The optimal concentration of this agent is about 2.4 mg/L with a contact time of at least one hour. This system can be used to decontaminate glove boxes, walk in incubators, and small rooms. An advantage of this system is that the end products (i.e., water) are not toxic. Low relative humidity can be used.<sup>33-36</sup>

### Chlorine Dioxide Gas

Chlorine dioxide gas sterilization can be used for decontamination of laboratory rooms, equipment, glove boxes, and incubators. The concentration of gas at the site of decontamination should be approximately 10 mg/L with contact time of one to two hours.

Chlorine dioxide possesses the bactericidal, virucidal, and sporicidal properties of chlorine, but unlike chlorine, does not lead to the formation of trihalomethanes or combine with ammonia to form chlorinated organic products (chloramines). The gas cannot be compressed and stored in high-pressure cylinders, but is generated upon demand using a column-based solid phase generation system. Gas is diluted to the use concentration, usually between 10 and 30 mg/L. Within reasonable limits, a chlorine dioxide gas generation system is unaffected by the size or location of the ultimate destination for the gas. Relative humidity does need to be controlled and high humidities are optimal. Although most often used in closed sterilizers, the destination enclosure for the chlorine dioxide gas does not need to be such a chamber. Because chlorine dioxide gas exits the generator at a modest positive pressure and flow rate, the enclosure also need not be evacuated and could be a sterility-testing isolator, a glove box or sealed BSC, or even a small room that could be sealed to prevent gas egress.<sup>40</sup> Chlorine dioxide gas is rapidly broken down by light; care must be taken to eliminate light sources in spaces to be decontaminated.

## B. Surface Decontamination

Liquid chemical germicides formulated as disinfectants may be used for decontamination of large areas. The usual procedure is to flood the area with a disinfectant for

periods up to several hours. This approach is messy and some common disinfectants represent a toxic hazard to laboratory staff. For example, most of the high-level disinfectants on the United States market are formulated to use on instruments and medical devices rather than on environmental surfaces. Intermediate and low-level disinfectants are formulated to use on fomites and environmental surfaces but lack the potency of a high-level disinfectants. For the most part, intermediate and low-level disinfectants can be safely used and, as with all EPA-registered disinfectants, the manufacturer's instructions should be closely followed.<sup>41</sup> Procedures for the use of chemical disinfectants should include safety precautions, the use of appropriate PPE, hazard communication, and training on spill response.

Some disinfectants that have been used for decontamination include sodium hypochlorite solutions at concentrations of 500 to 6000 parts per million (ppm); oxidative disinfectants (e.g., hydrogen peroxide, peracetic acid); phenols; and iodophors. Concentrations and exposure times vary depending on the formulation and the manufacturer's instructions for use. See Table 1 for a list of chemical disinfectants and their activity levels. A spill control plan must be available in the laboratory. This plan should include the rationale for selection of the disinfecting agent, the approach to its application, contact time and other parameters (e.g., if using a dilution, the frequency with which it will be replaced). Biological agents requiring BSL-3 and BSL-4 containment pose a high risk to workers and possibly to the environment, and these agents should be managed by trained, professional staff who are equipped to work with concentrated material.

Table 1: Activity Levels of Selected Liquid Germicides<sup>a</sup>

| Chemical                                 | Concentration                                       | Activity Level                                           |
|------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|
| Glutaraldehyde                           | Variable                                            | Sterilization; Intermediate- to high-level disinfection. |
| Ortho-phthalaldehyde (OPA)               | 0.55%                                               | High-level disinfection.                                 |
| Hydrogen Peroxide                        | 6-30%                                               | Sterilization                                            |
| Hydrogen Peroxide                        | 3-6%                                                | Intermediate- to high-level disinfection.                |
| Formaldehyde <sup>b</sup>                | 6-8%                                                | Sterilization                                            |
| Formaldehyde                             | 1-8%                                                | Intermediate- to high-level disinfection.                |
| Chlorine Dioxide                         | Variable                                            | Sterilization; High-level disinfection.                  |
| Peracetic Acid                           | 0.08%-0.23% with peroxide concentrations of 1-7.35% | Sterilization                                            |
| Peracetic Acid                           | Variable                                            | High-level disinfection.                                 |
| Hypochlorites <sup>c</sup>               | 500 to 6000 mg/L free/available chlorine            | Intermediate- to high-level disinfection.                |
| Alcohols (ethyl, isopropyl) <sup>d</sup> | 70%                                                 | Intermediate-level disinfection/                         |
| Phenolic Compounds                       | 0.5 to 3%                                           | Low- to intermediate-level disinfection.                 |
| Iodophors <sup>e</sup>                   | 30-50 mg/L free iodine                              | Low- to intermediate-level disinfection.                 |
| Quaternary Ammonium Compounds            | Variable                                            | Low-level disinfection.                                  |

a. This list of chemical disinfections centers on generic formulations. A large number of commercial products based on these generic components can be considered for use. Users should ensure that commercial formulations are registered with EPA or by the FDA. Users can search for EPA-registered products on the EPA's Pesticide Labels website.

b. Because formaldehyde is classified as a known human c carcinogen and has a low permissible exposure limit (PEL), the use of formaldehyde is limited to certain specific circumstances under carefully controlled conditions (e.g., for the disinfection of certain hemodialysis equipment). There are no FDA cleared liquid chemical sterilant/disinfectants that contain formaldehyde.

c. Generic disinfectants containing chlorine are available in liquid or solid form (e.g., sodium or calcium hypochlorite). The indicated concentrations are rapid-acting and broad-spectrum (i.e., tuberculocidal, bactericidal, fungicidal, and virucidal). Note: Common household bleach is an excellent and inexpensive source of sodium hypochlorite. Concentrations between 500 and 1000 ppm chlorine are appropriate for the majority of uses requiring an intermediate level of germicidal activity; higher concentrations are extremely corrosive as well as irritating to personnel, and their use should be limited to situations where there is an excessive amount of organic material or unusually high concentrations of microorganisms (e.g., spills of cultured material in the laboratory). In situations where there is an excessive amount of organic material present, the surfaces should be thoroughly cleaned to remove as much organic material as possible before applying sodium hypochlorite solution to disinfect the surface (see product label instructions). The concentration of the sodium hypochlorite should be determined in advance of use and the solution should be made fresh each day.

d. The effectiveness of alcohols as intermediate-level germicides is limited because they evaporate rapidly, resulting in short contact times, and also lack the ability to penetrate residual organic material. They are rapidly tuberculocidal, bactericidal and fungicidal, but may vary in spectrum of virucidal activity. **Items to be disinfected with alcohols should be carefully pre-cleaned then totally submerged for an appropriate exposure time.**

e. Only those iodophors registered with EPA as hard-surface disinfectants should be used, closely following the manufacturer's instructions regarding proper dilution and product stability. Antiseptic iodophors are not suitable to disinfect devices, environmental surfaces, or medical instruments

C. Chemical Safety

When using chemical agents for decontamination, pay attention to instructions for their use and Safety Data Sheets (SDS); ensure they are used safely and that appropriate precautions and protections are used. Exposures to disinfectants have resulted in occupational injuries such as cancer, hypersensitivities, dermatitis, and asthma.<sup>46,47</sup>

## D. Hand hygiene

Handwashing and hand decontamination are an underappreciated part of risk mitigation for handling pathogens. Gloves should be worn when handling biohazardous materials and hazardous chemicals, including those used in disinfection and decontamination; this does not replace the need for regular hand hygiene by laboratory personnel.<sup>48</sup> Hand hygiene should be performed after removing gloves, after touching potentially contaminated surfaces with bare hands, after completing work, and before exiting the laboratory. The main method of hand hygiene in the laboratory is handwashing with soap and water.

When handwashing facilities are not available, an alcohol-based hand sanitizer (ABHS) with an alcohol concentration between 60-95% may be used in conjunction with or in lieu of immediate handwashing, based on agent type and a risk assessment that accounts for potential reduced efficacy of hand sanitizers for soiled hands and inactivating some microorganisms (i.e., bacterial spores, parasites, and non-enveloped viruses). ABHS may be used for immediate hand hygiene until a handwashing facility can be accessed only if hands are not grossly contaminated. The limitations of ABHS should be communicated to staff. Handwashing with soap and water remains the preferred method of performing hand hygiene.<sup>49</sup> ABHS should be applied to cover the skin and nails (including underneath the nail) of the hands for 20-30 seconds. Posters are available to assist in demonstrating the proper method of hand sanitizing using ABHS on the CDC's Hand Hygiene website.

If hands are grossly contaminated when exiting the laboratory, they should be washed with soap or soap containing an antiseptic agent (i.e., antimicrobial soap) and water.<sup>49</sup> When using soap and water, the entire procedure should last 40-60 seconds from wetting hands to drying with a paper towel. Posters to assist in demonstrating the proper method of handwashing are available on the CDC's Handwashing website and on the World Health Organization's website.

## E. Transmissible Spongiform Encephalopathy Agents (Prions)

Prions are exceptionally difficult to inactivate and decontaminate and are the causative agent of Creutzfeldt-Jakob disease (CJD) and other transmissible spongiform encephalopathies of the central nervous system in humans or animals. Studies show that prions are resistant to conventional uses of heat and/or chemical germicides for the sterilization of instruments and devices.<sup>12,42,43</sup> Treatment of tissues and contaminated tissues is based on tissue infectivity.<sup>44</sup>

## F. Inactivation of Select Agents

Select agents can be inactivated using conventional disinfection and sterilization procedures appropriate to the type of agent (e.g., virus, spore-forming bacteria). Inactivation procedures typically leave cell components intact that can then be used as reagents for assay development or other studies while the purpose of disinfection is to kill and damage pathogens with no attention to preserve cell components. Once inactivated, the agents are no longer subject to the Select Agent Regulations. Problems have arisen when spore-forming Select Agents such as *Bacillus anthracis* have not been completely inactivated. This was highlighted in 2015 when irradiated spores were shipped to non-select, agent-approved laboratories but were later found to be only partially inactivated.<sup>45</sup> The Select Agent Regulations require that the inactivation process used for these agents be validated.

Select Agent guidance is available at:

- Guidance on the Inactivation or Removal of Select Agents and Toxins for Future Use; and
- Biosafety/Biocontainment Plan Guidance.

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## Section I: Biological Waste Disposal Policy

### Introduction

The UNC-Chapel Hill Biological Waste Disposal Policy stipulates proper procedures for the collection, decontamination, and disposal of laboratory-generated biohazard waste. This policy has been developed in order to minimize the risk of exposure to those who may come into contact with biohazard waste generated in a UNC-Chapel Hill research laboratory, specifically:

- lab workers generating and collecting biohazard waste during research,
- support staff retrieving, transferring, and autoclaving the biohazard waste,
- housekeeping staff responsible for transporting autoclaved waste in buildings that house UNC-Chapel Hill research laboratories,
- facilities staff (plumbers, electricians, HVAC, welders, etc), emergency personnel, and visitors who visit the lab infrequently,
- and employees responsible for hauling away waste that is generated in UNC-Chapel Hill research laboratories.

North Carolina medical waste rules (15A NCAC 13 B .1200), require that "Regulated Medical Waste", defined as "blood and body fluids in individual containers greater than 20 ml, microbiological waste, and pathological waste," must be treated before disposal in order to render the waste nonhazardous. Most UNC-Chapel Hill campus laboratory-generated biohazard waste, as defined below, falls under the State defined category of "microbiological waste" within 15A NCAC 13 B .1200. Biohazard waste generated and collected in UNC-Chapel Hill research laboratories is to be properly autoclaved according to procedures outlined below. This process changes the biological characteristics of the waste thereby reducing or eliminating its potential to cause disease. Laboratories with biohazard waste not specifically addressed by this document (such as waste with multiple hazards, e.g. radioactive biohazardous waste) should consult with Environment, Health and Safety (EHS) for alternative treatment and disposal methods.

The procedures of this policy are consistent with applicable sections of the OSHA Bloodborne Pathogens Standard 29 CFR 1910.1030.

## Defining Laboratory-Generated Biohazard Waste

All biohazard waste generated in UNC-Chapel Hill research laboratories will be properly treated prior to its transfer and final burial in the landfill or incineration. This biohazard waste includes:

- Materials contaminated or potentially contaminated during the manipulation or clean-up of material generated during research and/or teaching activities requiring biosafety level 1, 2, or 3 or animal or plant biosafety level 1, 2, or 3. Refer to your laboratory's Biological Hazards Registration section of the Laboratory Safety Plan (Schedule F) to identify these materials in your lab.
- Human liquid blood and body fluids.
- Human tissue and anatomical remains.
- Materials contaminated with human tissue or tissue cultures (primary and established) because these are handled at BSL-2.
- Animal carcasses, body parts, blood, fluids and bedding from animals infected with BSL2 and BSL3 agents.

Regulated Medical Waste Disposal Chart

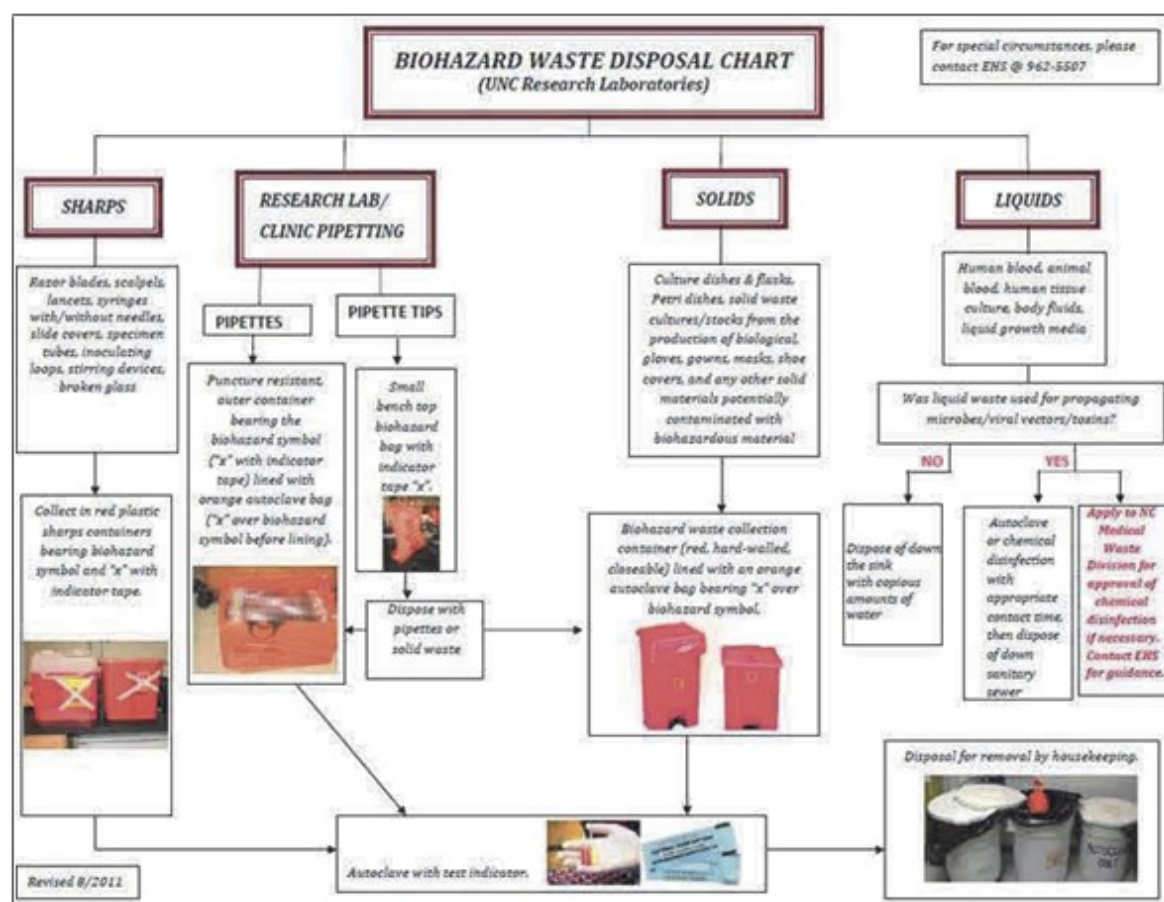
| Type of Regulated Medical Waste                                                                                                          | Disposal Method                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Blood and body fluids ( <b>Regulated medical waste</b> )                                                                                 | Treated with bleach or autoclaved and put down the sanitary sewer.                                                                                                                                                                                                                                                                                                                                |
| Microbiological Waste including Biosafety Level 1, 2 and 3 organisms: ( <b>Regulated medical waste</b> )                                 | Autoclaved at 121 °C. (Autoclaves are tested weekly using bio indicators <i>Geobacillus stearothermophilus</i> ) or chemically treated and put down sanitary sewer. <b>Please note:</b> The NC Medical Waste Rules do not allow chemical disinfection of regulated liquids followed by disposal to the sanitary sewer unless approval has been obtained from the NC Division of Waste Management. |
| Pathological waste (animal carcasses infected with human BSL1 and BSL2 this includes transgenic mice) ( <b>Regulated medical waste</b> ) | Animals are incinerated by vendor (Stericycle).                                                                                                                                                                                                                                                                                                                                                   |
| Pathological waste (animal carcasses infected with human BSL3 pathogens) ( <b>Regulated medical waste</b> )                              | Autoclaved in the lab at 121 °C. (Autoclaves are tested weekly using bio indicators <b><i>Geobacillus stearothermophilus</i></b> ). After autoclaving animals are incinerated by vendor (Stericycle).                                                                                                                                                                                             |
| Pathological waste (animal carcasses that were used for in vivo testing of pharmaceuticals) ( <b>Regulated medical waste</b> )           | Animals are incinerated by vendor (Stericycle).                                                                                                                                                                                                                                                                                                                                                   |
| Uninfected Animal Carcasses                                                                                                              | Animals are incinerated by vendor (Stericycle).                                                                                                                                                                                                                                                                                                                                                   |
| Non-hazardous Sharps                                                                                                                     | White plastic sharps containers sent to landfill.                                                                                                                                                                                                                                                                                                                                                 |
| Biohazardous Sharps                                                                                                                      | Red plastic sharps containers are autoclaved then sent to landfill.                                                                                                                                                                                                                                                                                                                               |

Biohazard waste originating from designated BSL-2 or greater containment areas must be indicated on the Biological Hazards Registration section of the Laboratory Safety Plan (Schedule F).

Refer to the chart below for a snapshot of the four most common biohazard waste collection methods. These methods are described in greater detail on subsequent pages. For contaminated animal carcasses, body parts, and bedding, refer to that particular section of this guide for disposal guidance.



## Biohazard Waste Disposal Chart



## Sharps Disposal Containers



## Biohazard Waste Collection Methods Contaminated Sharps

Include items such as:

- razor blades,
- scalpels,
- lancets,
- syringes with/without needles,
- slide covers, and
- specimen tubes

In UNC-Chapel Hill research laboratories, biohazardous sharps are collected directly into red, plastic containers available from FischerScientific(stock # 14830124 for contaminated, 1482664B for non-contaminated). These containers must bear the biohazard symbol marked with an "x" using autoclave indicator tape. Federal OSHA regulations (CFR 1910.1030) require biohazard laboratories to minimize their use of sharps whenever possible and that needles not be recapped, purposely bent, broken, or otherwise manipulated by hand. To avoid accidents related to overfilling the containers, remove the containers for disposal when they are 2/3 full. When removing the sharps container from a biosafety cabinet, always decontaminate the exterior of the container. Containers of sharps contaminated with biohazardous

materials should be autoclaved in an orange autoclavable bag marked with an "x" over the bag's biohazard symbol. After autoclaving, the bags with the containers of sharps can be disposed of with the regular trash. Non-hazardous sharps should be placed in the white plastic sharps containers. The non-hazardous sharps containers should be disposed of in regular trash once they are 2/3 full.

While small shards of contaminated broken glass can be placed into the sharps containers identified above, large contaminated broken glass items must be autoclaved separately in a hard-walled container (such as a cardboard box) lined with an orange biohazard bag bearing an autoclave tape indicator "x" over the bag's biohazard symbol. Place the tape on the orange bag *before* it is used to line the box to prevent contact with biohazardous materials and sharps. The universal biohazard symbol should also be found on the outside of the box. **After autoclaving**, the glass waste can be disposed of in the regular trash.

## Biohazardous Glass and Sharps



Non-Hazardous Sharps Box



Autoclave Bag



**DO NOT**  
enclose the  
cardboard

boxes used for gathering sharps/glass within an autoclave bag. This will prevent steam penetration during autoclaving. Steam penetration is crucial during the decontaminating process. Remember to line the boxes with an orange autoclave bag marked with an "x" over the biohazard symbol before lining the box

## Research Lab/Clinic Pipetting

For **large-scale** collection **outside** the biosafety cabinet of Glass (Pasteur) and plastic pipettes contaminated under the definition of biohazard waste, line a puncture-resistant outer container (such as the box the pipettes came in) with an orange autoclave bag marked with a heat sensitive autoclave tape "x" (available from FisherScientific as stock number #15-903) over the biohazard symbol. To avoid possible exposure, place the indicator tape "x" over the bag's biohazard symbol prior to loading the bag with pipettes. The universal biological hazard symbol must also be displayed on the outer container. When the box is full, close the inner bag leaving an opening for the steam to penetrate. Tape the outer box closed with autoclave tape. Do **not** use colored tape to close box.

## Inside the Biological Safety Cabinet



For frequently removed **small scale** collection, such as sterile pipetting in a biological safety cabinet, line a small orange autoclave bag inside a hard-walled collection container inside the cabinet. When the bag is 2/3 full, close it loosely, spray with proper disinfectant and transfer it to a larger scale pipette collection container located outside of the cabinet.



Another alternative for collecting biohazardous pipettes is to place them in a long, hard walled cylindrical container filled with an effective disinfectant. The pipettes should be allowed to remain in the disinfectant for the recommended contact time to ensure decontamination.



On the benchtop, pipette tips are to be collected in a small autoclave bag lining a wire stand or other container bearing the biohazard symbol. When 2/3 full, loosely close the bag to allow for steam penetration, spray with disinfectant and place with other solid biohazard waste.



## Contaminated Solids

Biohazard solids consist of:

- culture dishes, flasks,
- Petri dishes,
- solid waste cultures/stocks from the testing and production of biologicals,
- gloves, gowns, masks,
- and other solid material potentially contaminated under the definition of biohazard waste (above). The outer collection container must be durable, leak proof, have a lid and be of such a design so as not to be mistaken by Housekeeping as regular trash. This container must be labeled with a biohazard sticker. Wire cages cannot be used as the outer container.



In addition to the requirements that biohazard waste containers must be durable, leak-proof, have a lid, and be clearly labeled, **all UNC-Chapel Hill laboratories are required to collect biohazard waste in outer containers that are red.** This policy also puts a maximum limit on the size of a biohazard waste container at 15-gallons (57-L). All biohazard bags must be orange in color by June 2012. Fisher Scientific carries

orange autoclave bags that vary in size; contact the campus representative or for further details.

**The red biohazard container must be lined with an orange autoclavable biohazard bag. Before lining the container with the orange biohazard bag, crisscross the bag's biohazard symbol and/or markings with heat sensitive autoclave tape, (available from FisherScientific as stock number 15-903). The lid should be kept on the biohazard container when not in use. Remove bags at 2/3 full. Never place glass in these containers.**

## Liquids

Although the rules and definitions for liquid biohazard waste vary somewhat from solid waste procedures, autoclaving is the method of choice for disinfection of the following:

- Animal blood/body fluids from animals infected with BSL2 and BSL3 agents.
- Human tissue culture, human cell lines (primary or established)
- Human body fluids as defined under the UNC-Chapel Hill Laboratory Exposure Control Plan
- Liquid growth media removed from human tissue cultures

Autoclaved liquid wastes may be discharged directly to the sanitary sewer.

**Chemical disinfection** may be an acceptable alternative to autoclaving liquid biohazard waste generated in research laboratories at UNC-Chapel Hill such as bleach treatment. When this is done, care must be taken to avoid splash and the drains must be flushed with generous amounts of water. NC Medical Waste Rules do not allow chemical disinfection of regulated liquids followed by disposal to the sanitary sewer unless approval has been obtained from the NC Division of Waste Management.

**Regulated liquids** include the following:

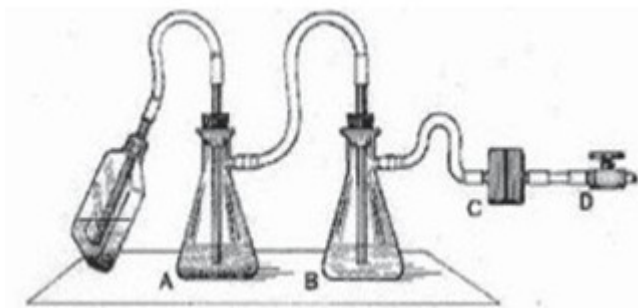
- Liquid waste media from cells/tissue used for propagating **risk group 1, 2, or 3 pathogens** or toxins, including those produced in recombinant DNA procedures.
- "Microbiological waste" as defined by the **North Carolina medical waste regulations**: e.g. cultures and stocks of infectious agents.
- from animals intentionally infected with microbes, viral vectors, or toxins

If you wish to obtain approval for chemical treatment of infectious liquids, you must provide information demonstrating the effectiveness of the chemical being used to treat the specific microbiological agents, taking into account factors such as temperature, contact time, pH, concentration, penetrability and reactivity of organic material. All requests for approval must be submitted to the NC Division of Waste Management through EHS, and documented in the Lab Safety Plan under Schedule F (Biological Hazards). Visit EHS's Chemical Treatment of Liquid Microbiological Waste webpage to evaluate if chemical treatment of your liquid biohazard waste requires approval.

Proper aspiration vacuum flask set up

- A - Primary flask –used to collect liquid
- B - Secondary flask (overflow flask) minimizes splash
- C - In line filter between secondary flask and vacuum source (FisherScientific stock number 09-744-75)
- D - Vacuum line that is occasionally serviced by lab workers or UNC-Chapel Hill support personnel





The primary and secondary flasks should contain a 10% bleach solution. The flask solution should be changed at least once a week to insure the killing strength of the bleach solution. Flask waste solution can be disposed of down the sink drain only after all potentially infectious material has had at least 20 minutes of contact time.

NOTE: If using a disinfectant other than a bleach solution, it may not be approved for sink disposal and you should contact the Biosafety Section at EHS (919-962-5507).

## Drosophila

An alternative to autoclaving *Drosophila* is dumping anesthetized flies directly into a container with a small amount of mineral oil or a bottle containing either ethanol or isopropanol. If you do not plan to re-use the material, these bottles must be labeled as ethanol, isopropanol or mineral oil waste to be picked up by EHS. If you are going to reuse the material you are dumping the *Drosophila* into, then you will label the bottle recycled ethanol, isopropanol etc. **These bottles of chemicals cannot be poured down the sink or sanitary sewer. They must be discarded using the online hazardous waste pick up program through EHS.**

## Contaminated Animal Carcasses, Body Parts, and Bedding

Animal carcasses are disposed of through the Department of Laboratory Animal Medicine. Animal carcasses, body parts, and bedding from animals inoculated with infectious agents, are disposed of by incineration. These materials are to be placed in boxes provided by the Division of Comparative Medicine (DCM; formerly the Division of Laboratory Animal Medicine (DLAM)) and marked for incineration. (No needles or other type of metal and no PVC plastic are to be placed in the collection boxes. Use only non-PVC plastic bags.) Carcasses contaminated with radioisotopes or carcinogens are picked up by the Department of Environment, Health and Safety.

### Human Tissues/Body Parts

Recognizable human anatomical remains or tissues and large tissues must be disposed of by incineration. Remains contaminated with hazardous chemical or radioactive substances require special disposal and EHS must be contacted for disposal.

Unrecognizable human tissues can be autoclaved and disposed of in regular trash. If the tissues have been chemically preserved, they can be disposed of as chemical hazardous waste.

## Loading and Unloading the Autoclave Safely

Contaminated materials should never be left in hallways or other public spaces prior to autoclaving. Biohazard bags should remain in the laboratory until they are ready to be placed in the autoclave. Never leave bags sitting on the floor next to the autoclave. Bags that are closed and ready for autoclaving must be placed in secondary containment as shown. If the bags are being transported to the autoclave, they must be contained in closed, hard-walled secondary containers.



Minimize contact with biohazard waste as much as possible. Never crush or push down biohazard waste. Biohazard waste containers should be removed for autoclaving when they

are 2/3 full. Indicator tape should be applied when placing the new autoclave bag into the hard walled outer container; this will reduce handling of the biohazard waste during removal. The heat sensitive autoclave tape should be placed in an "X" pattern over the biohazard symbol. The heat sensitive tape is to be of the type that changes color, such as the type that the word "autoclaved" appears after treatment. This tape is available from Fisher Scientific as stock number 15-903. Once the autoclave disinfection is complete, the tops of the bags may be sealed tightly with lab tape.

After the proper autoclave waste decontamination steps are followed, the decontaminated waste is then placed in a 44 gallon or 32 gallon white Rubbermaid Brute container (with a drum dolly), lined with black plastic garbage bags, and located in the vicinity of the autoclave. These containers are to be labeled "AUTOCLAVED/

DECONTAMINATED WASTE ONLY" (labels available on EHS website). Biohazard bags placed in the white Brute containers and marked with the heat sensitive tape signal to Housekeeping that the waste is safe and ready to be removed from the laboratory for disposal in the dumpster.



Each department is responsible for providing an adequate number of these containers which are available from Fisher Scientific. **Housekeeping will not remove or otherwise handle overflowing waste or waste in untreated biohazard bags.**



## Autoclaving Precautions

Autoclaving, or steam sterilization, is the most dependable procedure for the destruction of all forms of microbial life. Proper temperature and exposure time are critical factors in ensuring the reliability of this method. These critical factors are dependent upon steam penetration to every part of the waste load.

Therefore, the autoclave user must be mindful to prevent the entrapment of air. If all the air is not allowed to escape from the waste during the cycle, it cannot be replaced by steam. Saturated steam is employed under pressure (at least 15 pounds per square inch) to achieve a chamber temperature of at least 121 C (250 F) for a minimum of 15 minutes. This time is measured **after** the temperature of the steam saturated material being sterilized reaches 121 C.

The hazards associated with autoclaves include extreme heat and high pressure and large, heavy doors and loading carriage. When operating an autoclave the following safety procedures must be followed:

1. Become familiar with the autoclave's owner's manual. Though the principle is the same for each, manufacturer recommendations for use can vary widely.
2. Firmly lock autoclave doors and gaskets in place before you run the autoclave to prevent a sudden release of high-pressure steam. Some autoclaves do not have safety interlocks that prevent the autoclave from running if the door isn't closed properly. If your autoclave does not have safety interlocks, you will need to take additional precautions to ensure that the doors are closed.
3. If you have an older autoclave that has little or no heat shielding around the outside, attach signs warning of "Hot Surfaces, Keep Away" on or next to the autoclave to remind people of the hazard. Do not stack or store combustible materials (cardboard, plastic, volatile or flammable liquids, compressed gas cylinders) next to an autoclave.
4. Do not autoclave toxic, volatile or radioactive material. If you have biohazard waste that contains any of these materials, please contact EHS for guidance.
5. When a cycle is complete, wait approximately 1-2 minutes after the pressure gauge reads zero before opening the door of the autoclave.
6. Wait at least 30 seconds after opening the door before reaching or looking into the autoclave.
7. Open the door slowly, keeping head, face, and hands away from the opening.
8. Allow contents to cool before removing them from the autoclave.
9. Remove solutions from the autoclave slowly and gently; some solutions can boil over when moved or when exposed to room temperature. Thick, heat-resistant gloves, safety goggles or face shield and a rubber apron must be worn when removing hot liquids from the autoclave. Liquids should stand for over 1 hour before being handled without heat-resistant gloves.
10. Clean up any spills immediately.
11. Report any malfunctions or accidents immediately to your supervisor.

## Training

All employees that use an autoclave must complete the online autoclave training. To ensure that infrequent users do not neglect proper operating techniques, autoclave operating instructions should be posted in close proximity to the autoclave.

## Autoclave Waste Decontamination Procedures

- The autoclave is to be operated at 121°C (250°F) or higher for a minimum of 60 minutes for most biohazard waste (see chart below). The time and temperature used for each type of waste in the laboratory must be validated using biological indicators to ensure effective sterilization (see procedure below). Some autoclaves are equipped to operate at higher temperatures, which would allow for shorter exposure times.

Criteria for Autoclaving Typical Materials

| Material                                           | Temperature                | Time       |
|----------------------------------------------------|----------------------------|------------|
| Laundry                                            | 121 C (250 F)              | 30 minutes |
| Trash (biohazard bags containing infectious waste) | 121 C (250 F)              | 1 hour     |
| Glassware                                          | 121 C (250 F)              | 1 hour     |
| Liquids                                            | 121 C (250 F), each gallon | 1 hour     |
| Animals                                            | 121 C (250 F)              | 8 hours    |

- Use the appropriate autoclave settings. Autoclaves may have settings for "LIQUIDS" to be used for liquid materials. "LIQUID" settings run for longer periods at lower temperatures to minimize liquid evaporation and spills. For solid materials, the "DRY GOODS WITH VACUUM" should be used for infectious waste as it is the most effective at moving steam and heat into the deepest parts of large bags producing the best conditions for killing persistent organisms. "DRY GOODS WITHOUT VACUUM" should only be used for clean items that need to be sterilized. Exhaust settings should also be appropriate for the type of waste being autoclaved. FAST exhaust should be used for solid items and SLOW exhaust should be used for liquids.
- Solid waste.** Do not overfill waste bags or the autoclave. This will interfere with steam penetration. Add about 50-100 ml (~1/4 to 1/2 cup) of water to each bag of solid waste to facilitate steam penetration in the bag. If there is naturally occurring water in the load, adding additional water is not necessary. Keep the waste bags slightly open to allow for steam penetration. Bags are placed into stainless steel or polypropylene trays prior to autoclaving.
- Liquid waste.** Liquids should be placed in borosilicate (Kimax or Pyrex) or polypropylene containers for autoclaving. The containers should not be filled to more than 75% capacity. The caps or stoppers on the containers should be loosened. **Never autoclave sealed containers of liquid. This could result in an explosion of super heated liquid.** Liquid containers should be placed in a stainless steel or polypropylene tray with ¼ to ½ inch of water in the bottom of the tray. The tray should be placed on a shelf in the autoclave and not on the bottom of the chamber.
- N. C. medical waste rules state that autoclaves are to be provided with a chart recorder which accurately records time and temperature for each cycle.

## Autoclave Waste Decontamination Cycle Testing & Verification

- The N.C. Medical Waste Rules require that autoclaves be monitored under conditions of full loading for effectiveness weekly through the use of **biological indicators**. *Geobacillus stearothermophilus* indicators must be used with average spore populations of 10<sup>4</sup> to 10<sup>6</sup> organisms. There are many commercially available biological indicators with a choice of spore ampoules or spore strips with growth media.



2. Follow the instructions provided by the

- manufacturer of the biological indicators. Most require refrigeration when kept in storage.
- Place the indicator in the middle of the waste bag or material to be autoclaved. It is best to put the indicator in the waste bag before it is filled completely. To aid recovery of the indicator after sterilization, tape it to a brightly colored sheet of paper or to a long string allowed to protrude from the bag. Indicators can also be placed in test waste bags filled with materials that simulate full loading for the test.
- Autoclave the waste following normal procedures. Once the cycle is complete and contents have cooled, remove the indicator from the waste bags wearing appropriate protective equipment. Prepare and incubate the indicator and a control indicator that was not autoclaved as recommended by the manufacturer.
- Check for signs of growth at regular intervals during the incubation period (8, 12, 24 and 48 hours). There should be signs of growth on the control indicator that was not autoclaved or the test is invalid. If there are signs of growth on the indicator placed in the waste, the waste was not sterilized properly. The time, temperature and autoclave procedures should be re-evaluated. If an autoclave problem is suspected, Facilities Services must be contacted immediately for

repair.

6. A log of each test should be maintained, which includes the type of indicator used, date, time, and result of the test. An autoclave testing log is available for download at the EHS website.
7. The waste does not have to be held until the results of the testing confirm effectiveness. If test results indicate that the autoclave is not sterilizing properly, the autoclave should not be used for waste until it has been repaired. The first load run in the autoclave should be tested with a biological indicator to insure proper functioning of the autoclave.

## Autoclave Preventative Maintenance

Autoclave operators should perform the following preventative maintenance on their autoclave to maintain the autoclave's effectiveness:

1. Remove the plug screen or drain strainer to make sure it is free of dirt, dust, or sediment that may collect in it and it should be cleaned as necessary.
2. Clean the interior surfaces of residues collected from the steam or materials being sterilized as needed.
3. Visually inspect the gaskets, doors, shelves and walls for residue buildup or wear regularly.
4. Report any problems with your autoclave to Facilities Services.

[Back to Chapter Nine](#)

[Proceed to Chapter Eleven](#)

## Attachments

### Attachments (1)

Biohazard Waste Disposal Chart.pdf

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